

Abstract No.

Cryo-mtscATAC-seq resolves region-specific myeloid clonal architecture in human glioblastoma

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High-throughput clonal tracing of primary human samples relies on naturally occurring barcodes, such as somatic mitochondrial DNA (mtDNA) mutations detected via single-cell ATAC-seq (mtscATAC-seq). Fresh-frozen clinical specimens preserve tissue architecture but compromise cell integrity, thereby precluding their use in multi-omic approaches such as mitochondrial genotyping at single-cell resolution. Here, we introduce Cryo-mtscATAC-seq, a broadly applicable method for diverse pathophysiological contexts to isolate nuclei with their associated mitochondria ("CryoCells") from frozen samples for high-throughput clonal analysis, validated across neurodegenerated brain, pediatric neuroblastoma and human aorta. In glioblastoma (GBM), used here as a model to benchmark clonality, we profiled matched primary and recurrent tumors and recovered clonally informative mtDNA variants shared along the longitudinal axis. Focussing clonal tracing to the immune compartment, myeloid-specific mtDNA variants defined tumor-associated macrophage (TAM) clonotypes. These myeloid clones were distributed in a region-specific manner across tumor sections, with clonal composition shifting between regions. Individual TAM clones shared broad phenotype but mapped onto discrete functional states, linking clonal lineage to myeloid state. Together, Cryo-mtscATAC-seq extends mtDNA genotyping to archival frozen GBM specimens, resolving clonal evolution and region-specific myeloid clonality within a single assay — opening new avenues for clonal tracing in human tissue.



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